

Aysha Al Rashid

Agentic AI & Data engineer Developer | Software Developer

✉ aysharashid.tech@gmail.com ☎ +971544291807 📍 Abu Dhabi, United Arab Emirates

🌐 <https://github.com/Aysha-Rashid> 🔗 [linkedin.com/in/aysha-al-rashid](https://www.linkedin.com/in/aysha-al-rashid)

Summary

Software developer with hands-on experience rapidly prototyping Agentic AI workflows, document-grounded assistants, data validation pipelines, Geospatial analytics dashboard platforms and backend services. Skilled in Python, C/C++, TypeScript, FastAPI, PostgreSQL/PostGIS, LangGraph, Pandas, SQL and Agentic AI workflows. Experienced in developing projects from system design and data processing through API development, user interfaces, testing, and deployment. Available for six-month hybrid internship.

Education

Bachelor in Software Development and Artificial Intelligence, 42 Abu Dhabi 10/2025 – Present | Abu Dhabi
Focus: AI systems, data management, backend architecture, High-level programming, and applied software engineering.

Diploma in Software Development, 42 Abu Dhabi 07/2023 – 10/2025 | Abu Dhabi
Coursework and projects: C, C++, systems administration, Docker, networking, full-stack web development, and software architecture.

Projects

AI Infographic Generation Platform, Full-Stack AI Software Development Project

Technologies: Python, FastAPI, Pydantic, LLM APIs, Generative AI, Image Generation APIs, REST APIs, HTML/CSS/JavaScript, Docker

- Built an end-to-end **Generative AI application** that converts natural-language prompts into structured, editable infographics using LLMs, image generation, and deterministic rendering.
- Designed an AI pipeline for **content planning, semantic validation, visualization selection, asset generation, layout composition, and rendering.**
- Implemented **Pydantic validation and retry workflows** to detect invalid structures, duplicate content, unsupported statistics, and unreliable LLM outputs.
- Developed a **scene-based rendering architecture** with editable text/image layers, separating generative AI decisions from deterministic layout and validation logic.

AI-Assisted Research and Knowledge Application

Technologies: React, JavaScript, Tailwind CSS, Gemini API, Gemini Canvas, PDF.js, Lucide React, Vercel

- Built a document-grounded knowledge assistant that extracts content from uploaded research papers and supports question answering, summarization and information discovery.
- Integrated the Gemini API to answer document-based questions, summarize research content, and support faster information discovery.
- Developed features for generating presentation slides, visual mind maps, study notes, and AI-generated quizzes from uploaded documents.
- Used PDF.js to extract and process PDF content and React with Tailwind CSS to create a responsive, user-friendly interface.

Agentic AI Crime Intelligence & Geospatial Risk Dashboard, Full-Stack AI and Geospatial Software Project

Technologies: React, Leaflet, FastAPI, PostgreSQL, PostGIS, Docker, Python, LLM APIs

- Developed an AI ingestion workflow that extracts incident category, location, date, severity, confidence and operational summaries from unstructured reports.
- Validated and stored structured outputs in PostgreSQL/PostGIS for filtering, proximity search and risk-based prioritization.
- Created an interactive dashboard that communicates incident patterns and nearby risks to non-technical users.
- Documented data sources, extraction assumptions and confidence values to support human review.

Full-Stack Multiplayer Pong Web Application, Team Lead & Full-Stack Developer

Technologies: JavaScript/TypeScript, REST APIs, WebSockets, Docker, Linux

- Led a **4-person development team** in designing and building a secure full-stack multiplayer Pong platform with user authentication, real-time gameplay, chat, player matchmaking, and multiplayer session management.
- **Implemented the initial authentication logic** and contributed to backend authentication workflows, including user sessions and frontend-backend integration.
- Containerized application services with **Docker** and managed development and deployment workflows in a Linux environment, improving consistency across team members and application components.

Technical Skills

Backend and Full-Stack

FastAPI, Node.js, Fastify, React, Streamlit, REST APIs, WebSockets, Git/Github, testing, clean code practices, authentication, frontend-backend integration

AI and Agentic Systems

LLM workflows, LangGraph, Gemini API, prompt engineering, intent classification, sentiment analysis, embeddings, vector databases, human-in-the-loop workflows

Data Engineering and Analytics

Python, Pandas, GeoJSON, GeoPandas, PostgreSQL, PostGIS, NumPy, Matplotlib, Tableau, ETL/ELT pipelines, data ingestion, data validation, data transformation, data cleaning, exploratory data analysis, Leaflet, spatial queries, hotspot analysis, map visualization

DevOps and Cloud

Docker, Docker Compose, NGINX, Linux, CI/CD fundamentals

AI-Assisted Development

Gemini, ChatGPT, Claude code, Grok AI, lovable, Gemini Canvas.

Courses

Agent AI, Hugging Face LLM workflows, AI agent design, tool integration, multi-agent orchestration, and autonomous task execution.	05/2026 – Present
--	-------------------

Piscine for Data Science, 42 Abu Dhabi Python-based data processing, PostgreSQL, data ingestion, ETL/ELT pipelines, data transformation, and data visualization.	02/2026 – 08/2026
--	-------------------

Associate Data Engineering, DataCamp SQL, relational data modeling, ETL pipeline development, data warehousing, database management, and cloud data fundamentals.	03/2026 – 05/2026
---	-------------------

Master Data Analyst Program, Simplilearn Data wrangling, SQL queries, Python-based data exploration, visualization, and business intelligence dashboards.	2022 – 2024
---	-------------

Achievements

2nd Place Winner – Athar+ Social Impact Hackathon, Athar+
Designed an AI-assisted platform to connect older adults with local community activities, demonstrating application of technology for public benefit.

Mentorship – ADEK Charter Schools
Mentored students in Python programming and logical problem-solving for Discovery Piscine; guided PISA readiness programs introducing AI concepts through hands-on activities.